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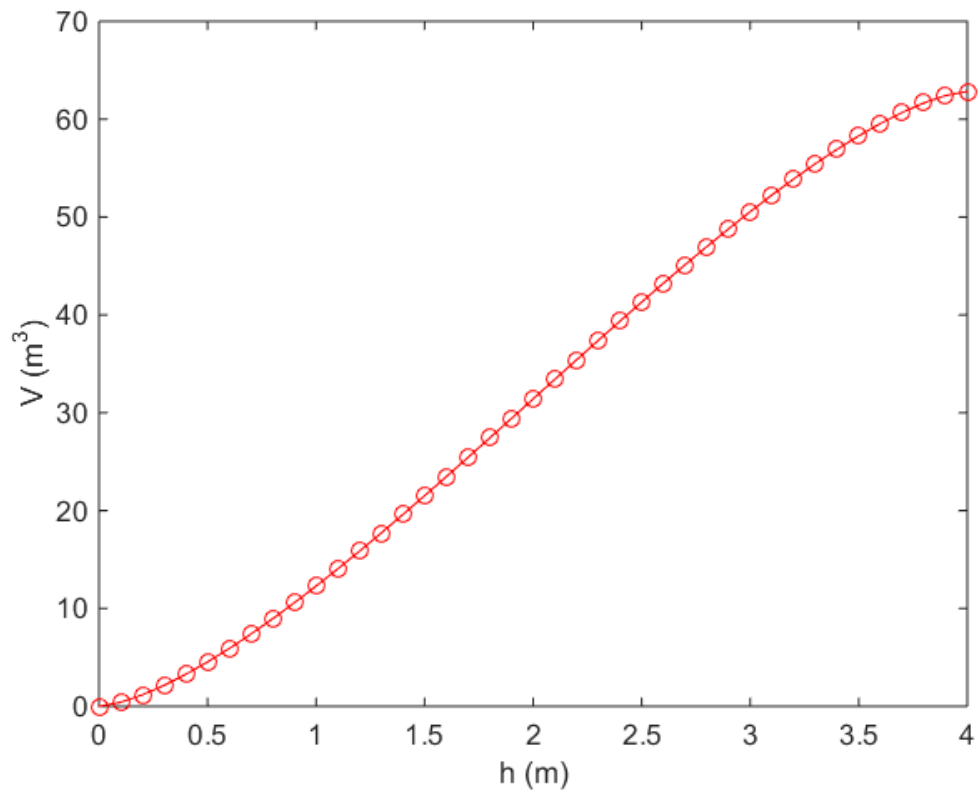
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CE 536 Assignment 1 Solutions

```
function assignment1
ass1_1;
ass1_2;
ass1_3;
ass1_4;
ass1_5;
ass1_6;
ass1_7;
end
```

Problem 1

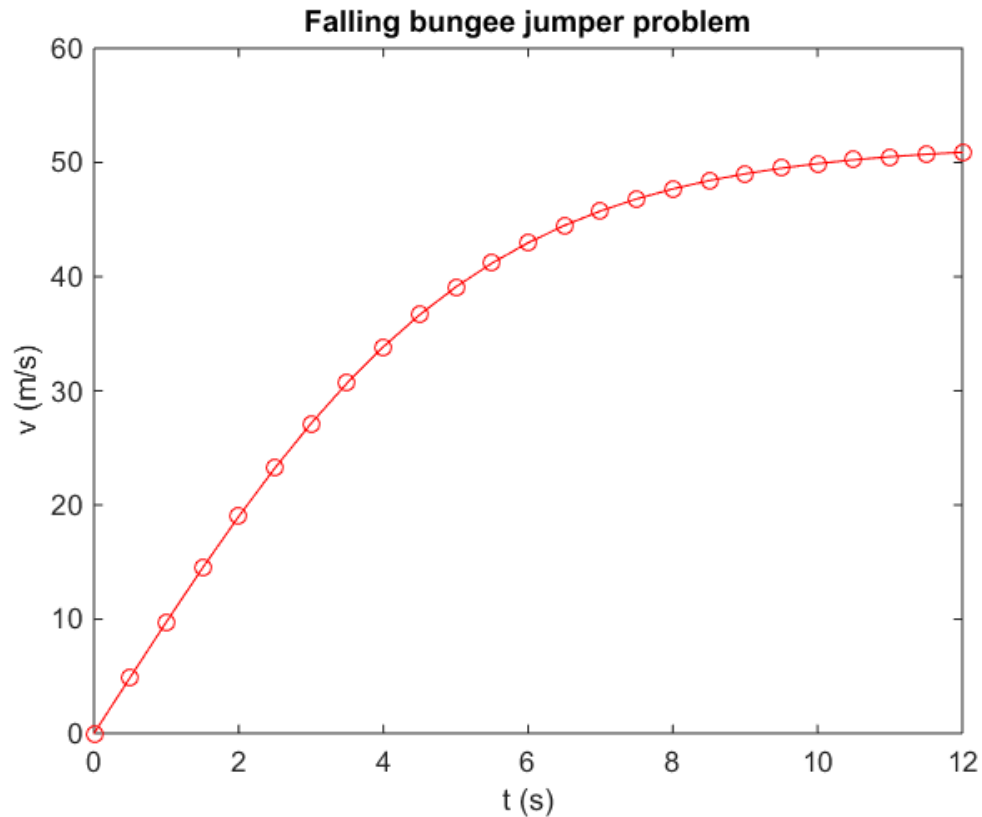
```
function ass1_1
r=2; L=5;
h=0:0.1:2*r;
V=(r^2*acos((r-h)/r)-(r-h).*sqrt(2*r*h-h.^2))*L;
figure(1);
plot(h,V,'-ro');
xlabel('h (m)');
ylabel('V (m^3)');
hold off;
end
```



Problem 2

```
%parachute problem
%solve  $dv/dt = g - cv^2/m$ 
function ass1_2
[t, v]=euler_bungee;
figure(2);
plot(t, v, '-ro');
xlabel('t (s)');
ylabel('v (m/s)');
title('Falling bungee jumper problem');
hold off;
end

function [t, vel] = euler_bungee
% solve numerically using Euler's method
deltat=0.5;tfinal=12;
g=9.8; m=68.1; c=0.25;
t=0:deltat:tfinal;
n=length(t);
vel=zeros(1,n);
for i=1:n-1
    vel(i+1)=vel(i)+(g-c*vel(i)^2/m)*deltat;
end
end
```



Problem 3

bioreactor problem with Monod kinetics

```
function ass1_3
[t_anal, c_anal] = monod_analytical;
[t_num, c_num] = monod_numerical;
figure(3);
plot(t_anal,c_anal,'-ro', t_num, c_num, '-b')
legend('analytical','numerical');
xlabel('t (days)');
ylabel('c (mg/L)');
title('Bioreactor Problem with Monod Kinetics');
hold off
end

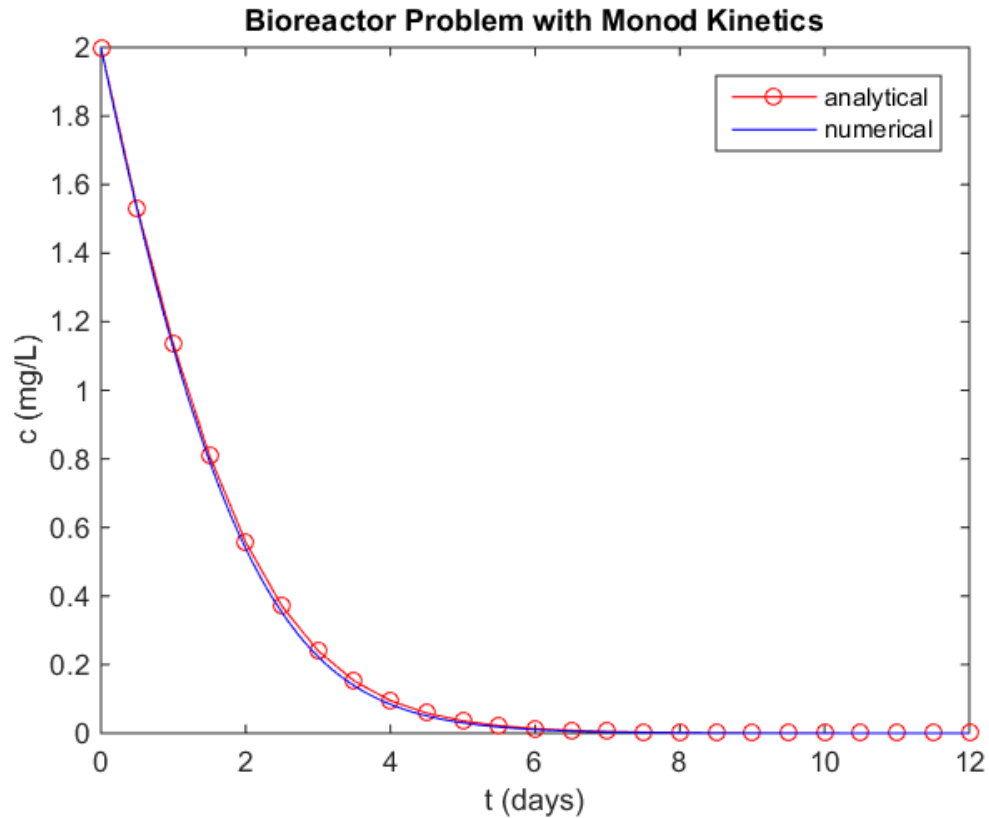
%analytical solution using dsolve
function [t, c] = monod_analytical
clear
syms c t
c=dsolve('Dc=-2*c/(2+c)', 'c(0)=2');
t=0:0.5:12;
c=subs(c);
end

function [t, c] = monod_numerical
```

```

%numerical solution
K=2; c0=2; deltat=0.1; mu=4; X=0.5;
t=0:deltat:12;
n=length(t);
c=zeros(1,n);
c(1)=c0;
for i=1:n-1
    c(i+1)=c(i)-(mu*X*c(i)/(K+c(i)))*deltat;
end
end

```



Problem 4

```

%storage tank problem
function ass1_4
[t, y]=euler_storage_tank;
figure(4);
plot(t, y, '-ro');
xlabel('t (s)');
ylabel('y (m)');
title('Storage tank problem');
hold off;
end

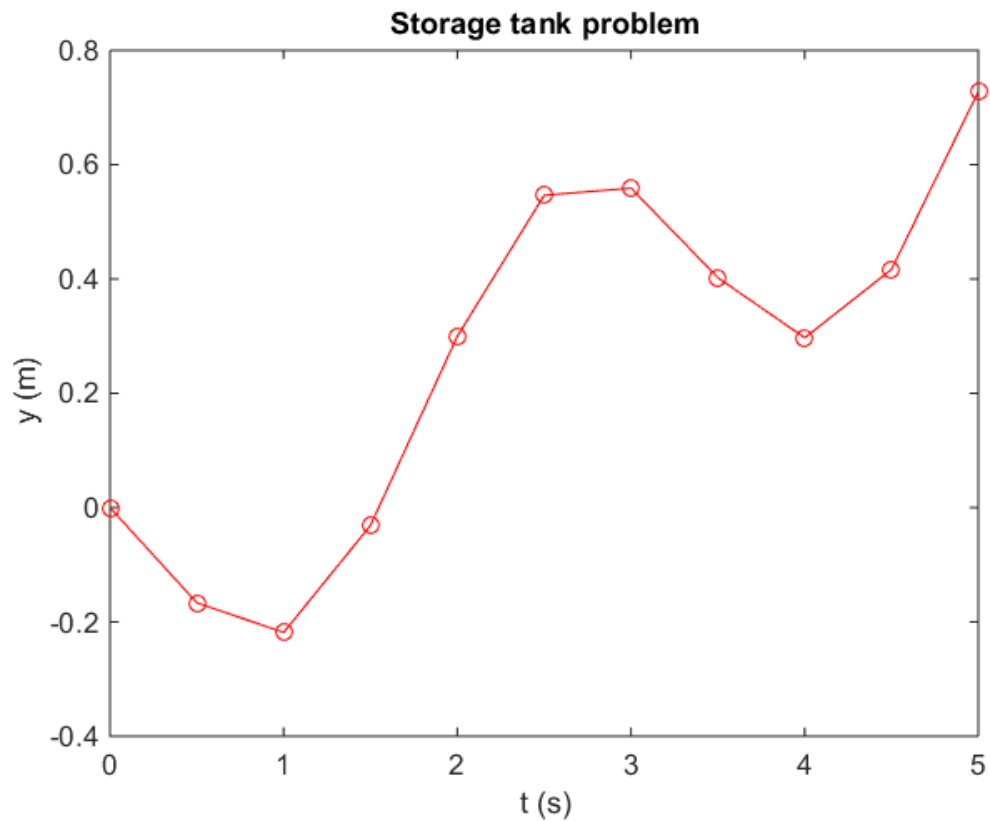
% solve numerically using Euler's method
function [t, y] = euler_storage_tank

```

```

deltat=0.5;tfinal=5;
Q=400; A=1200;
t=0:deltat:tfinal;
n=length(t);
y=zeros(1,n);
for i=1:n-1
    y(i+1)=y(i)+Q*(3*(sin(t(i)))^2-1)*deltat/A;
end
end

```



Problem 5

chemical reactor problem

```

function ass1_5
% define matrix
Q=[6 0 -1 0 0;
    1 -1 0 0 0;
    0 -1 9 0 0;
    0 1 8 -11 2;
    -3 -1 0 0 4];
% define rhs
b=[50; 0; 160; 0; 0];
% solve
C=Q\b
end

```

$C =$

11.5094
11.5094
19.0566
16.9983
11.5094

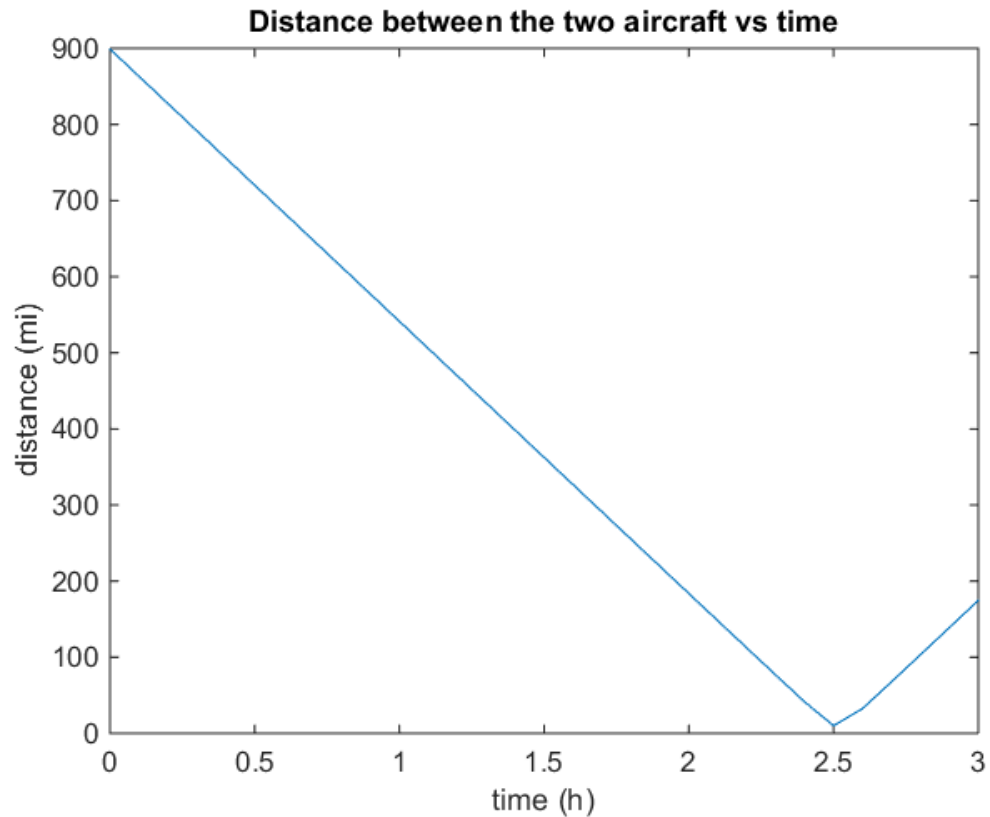
Problem 6

moving aircraft problem

```
function ass1_6

% define constants
t=0:0.1:3;
v1=320; v2=160;
d1=800-v1*t;
d2=410-v2*t;
% distance between aircraft
D=sqrt(d1.^2+d2.^2);

figure(6);
plot(t,D);
xlabel('time (h)');
ylabel('distance (mi)');
title('Distance between the two aircraft vs time');
hold off;
```



calculate solution using fzero

```
d30=fzero(@dist,0);
fprintf('Time at which the two aircraft are within 30 mi of each other = %g h\n',d
```

Time at which the two aircraft are within 30 mi of each other = 2.43246 h

```
end
```

```
% function returns D-30 for input into fzero
```

```
function [f] = dist(t)
```

```
v1=320; v2=160;
```

```
d1=800-v1*t;
```

```
d2=410-v2*t;
```

```
D=sqrt(d1^2+d2^2);
```

```
f=D-30;
```

```
end
```

Problem 7

```
function ass1_7
```

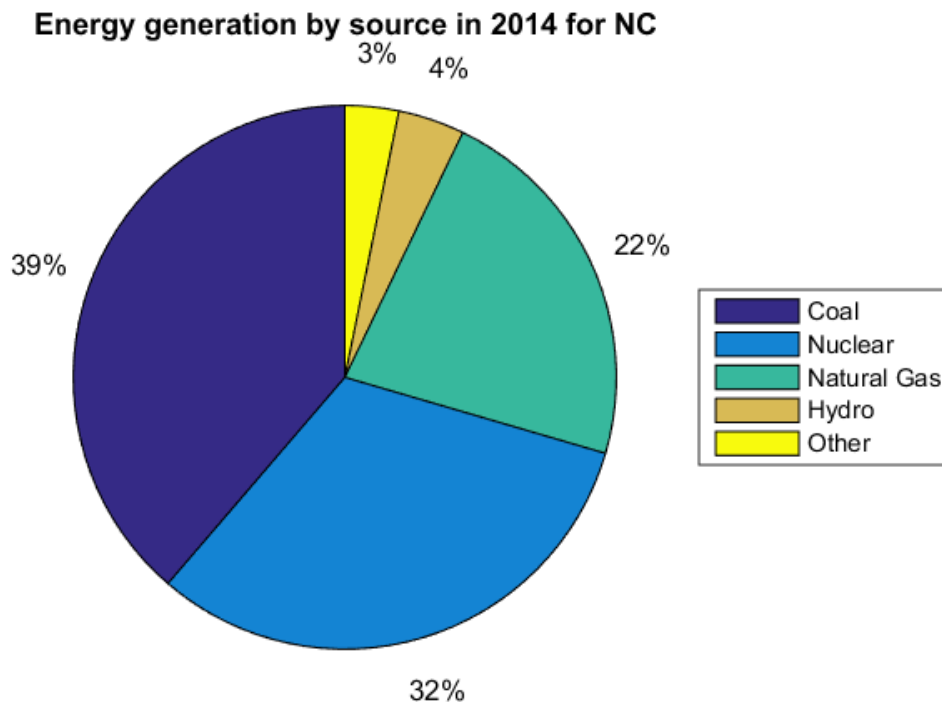
```
ass1_7a;
```

```
ass1_7b;
```

```
end
```

Part I - North Carolina

```
function ass1_7a
[num,txt,A]=xlsread('energy_generation_nc.xlsx','B6:C10');
nrows = size(A,1);
for i=1:nrows
    labels{i} = A{i,2};
    values(i) = A{i,1};
end
% plot results
figure(1);
pie(values);
legend(labels,'Location','eastoutside','Orientation','vertical');
title(['Energy generation by source in 2014 for NC']);
end
```



Part II - Any state

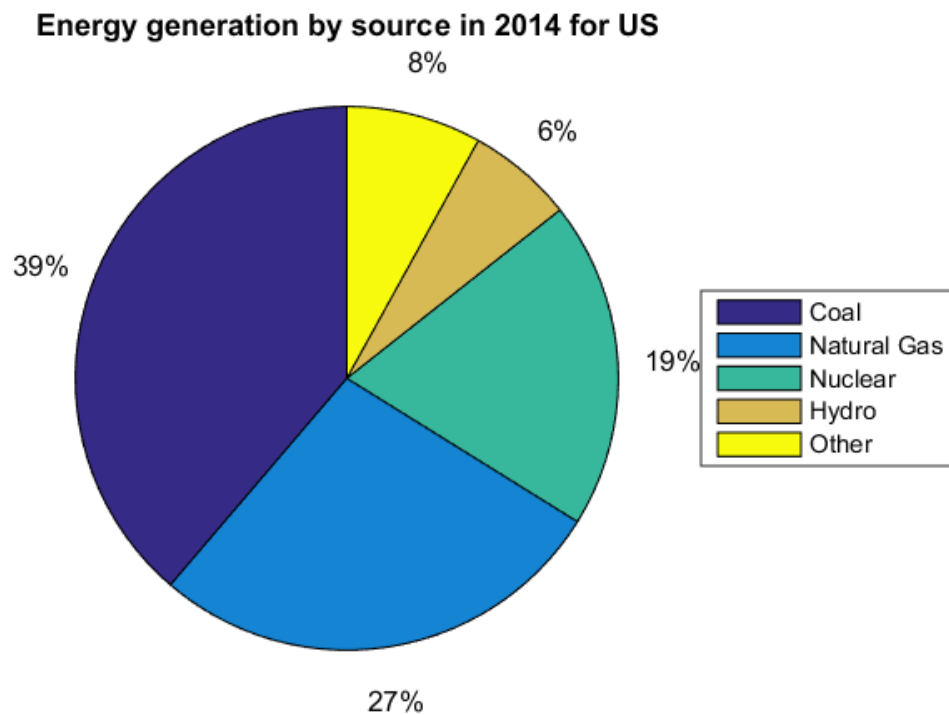
plots energy data from excel file for a given state

```
function ass1_7b
[num,txt,A]=xlsread('energy_generation_all_states.xlsx');
% state=input('Enter the two letter state abrevation (e.g., NC)\n','s');
state='US';
% extract data for given state
```

```

k=0;
nrows = size(A,1);
for i=1:nrows
    if (strcmpi(A{i,1},state) == 1)
        k = k +1;
        labels{k} = A{i,2};
        values(k) = A{i,3};
    end
end
% plot results
figure(2);
pie(values);
legend(labels, 'Location', 'eastoutside', 'Orientation', 'vertical');
title(['Energy generation by source in 2014 for ',state]);
end

```



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